Detecting and Attributing AI-Generated Images: A Forensic Fingerprinting Study

A Report Submitted
in Partial Fulfillment of the Requirements
for the Degree of
Bachelor of Technology
in
Computer Science & Engineering

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ALLAHABAD PRAYAGRAJ
October, 2026

UNDERTAKING

I declare that the work presented in this report titled "*Detecting and Attributing AI-Generated Images: A Forensic Fingerprinting Study*", submitted to the Computer Science and Engineering Department, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, for the award of the ***Bachelor of Technology*** degree in ***Computer Science & Engineering***, is my original work. I have not plagiarized or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, I accept that my degree may be unconditionally withdrawn.

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CERTIFICATE

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Abstract

AI image generators now produce photorealistic images, which raises a forensic question: was an image captured by a camera, or produced by a generative model, and if so, by which one? This project treats AI-generated image detection as a *source identification* problem, in which the generator plays the role that a camera plays in classical media forensics. We compare three generations of detectors (a fine-tuned ResNet-50, frozen CLIP features with a linear classifier and the NPR forensic residual) on known and unseen generators, under JPEG compression, resizing and blur, and for attribution to a specific generator, and then test them on current generators and on images shared through WhatsApp.

Careful data preparation proved essential. File properties alone separated our pilot images perfectly, and a detector trained on a popular benchmark without correction lost 19.9 percentage points (98.8% to 78.9%) once the AI images were compressed like the real ones. On de-biased data all detectors exceeded 96% accuracy and attribution to one of seven sources reached 86.1%. Detectors trained on 2021–2023 generators fell to a ROC-AUC of 0.72–0.75 on 2024–2026 generators and close to chance after WhatsApp sharing, while resizing and blur made real photographs look generated. Training on recent generators and analysing the whole image restored a ROC-AUC of 0.93 after sharing. A detector with a ROC-AUC of 0.999 on a benchmark still called most real phone and camera photographs AI-generated; adding real-world photographs from 57 phone models fixed this, and the final detector called 98% of Android and iPhone photographs real while detecting 97% of ChatGPT and 99% of Gemini images. Our website combines this detector with a check of signed Content Credentials, which identified images downloaded from ChatGPT and Gemini with certainty but are removed by sharing. On 62 files

we generated ourselves with ChatGPT, Gemini and Grok, including screenshots and WhatsApp copies, the website called 92% AI-generated. Attribution of current generators reached 67% on benchmark images but did not transfer to these files.

Acknowledgements

We thank our mentor, Dr. Anuja Dixit, for guidance throughout this project, and the Department of Computer Science and Engineering, MNNIT Allahabad, for its support.

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Chapter 1

Introduction

1.1 Motivation

Online platforms, newsrooms and courts increasingly need to know where an image came from. Generative models such as diffusion models now produce images that are hard to tell apart from photographs by eye.

Metadata cannot answer this question reliably. In our pilot data, every real iPhone photo carried 11 EXIF tags naming the camera, while every AI image carried none. A screenshot or a messaging app strips such metadata from a real photo, and an AI image can be re-saved with any metadata at all. Active approaches, such as watermarks or signed provenance records, work only when the creator cooperates and the record survives sharing. This project therefore follows the *passive forensics* approach: it looks for traces that the image-creation process leaves in the pixel data unintentionally [13].

1.2 Problem Statement

Given only the pixels of an image, decide whether it was produced by a camera or by a generative model and, where possible, identify the generator. The decision should remain reliable for generators not seen during training and after common image processing.

1.3 Research Questions

RQ	Question	Classical forensics analogue
RQ1	Can a detector separate real photographs from AI-generated images from known generators?	Was this image captured by a camera?
RQ2	Does it still work on generators never seen in training?	A camera missing from the reference database
RQ3	How does performance change under JPEG compression, resizing and blur?	Camera fingerprints weakening after processing
RQ4	Can it identify *which* generator produced an image?	Camera model identification

Table 1: Research questions mapped to classical source identification.

1.4 Contributions

- A de-biasing procedure for Tiny GenImage that matches real and generated images in format, resolution, colour mode and JPEG compression, including quality-factor matching and grid-aligned cropping.
- A measurement of shortcut learning: the same detector trained on the original (biased) and on the de-biased data, evaluated on the same test set.
- A comparison of three generations of detectors (a fine-tuned CNN, frozen CLIP features and the NPR forensic residual) and a simple combination of the last two, on known and unseen generators and under common image processing.
- A "then versus now" study: detectors trained on 2021–2023 generators tested on 2025–2026 generators, and, conversely, detectors trained on current generators.
- A real-world sharing test in which real and generated images are sent through WhatsApp before detection.

- Visualisations of generator fingerprints in the frequency domain and in the NPR residual.
- A deployment detector trained on current generators and on real photographs from 57 phone models, and a measurement of how a near-perfect benchmark score can hide a real-world failure.
- An attribution model for nine current generator families, a map of the feature space, and a test on 62 files generated by us with ChatGPT, Gemini and Grok, including screenshots and WhatsApp copies.
- A website that combines a check of signed Content Credentials (C2PA) with full-image pixel analysis, and shows the evidence behind each verdict.

Chapter 2

Background and Related Work

2.1 Digital Image Forensics

Every step that creates or processes an image leaves statistical traces. A *trace* is any such footprint; a *fingerprint* is a trace that is consistent across all images from one source, distinctive between sources and reasonably stable under normal processing. In a camera, the lens, colour filter array, sensor, demosaicing, in-camera processing and JPEG compression each leave characteristic traces.

The best-known fingerprint is the photo-response non-uniformity (PRNU) of the camera sensor. Lukáš *et al.* [6] extract a noise residual (an image minus its denoised version), average residuals over many images from one camera to obtain a reference pattern and identify the camera of a new image by correlation.

2.2 Generator Fingerprints

A fully AI-generated image never passes through a lens or sensor. Instead, the generator's architecture and training can leave traces of their own. Marra *et al.* [7] showed that GANs leave model-specific fingerprints, in direct analogy with camera PRNU. This analogy is the basis of the project: detecting and attributing AI images is source identification in which the source may be a generative model.

2.3 Learned Detectors

Wang *et al.* [14] trained a standard image classifier (ResNet-50) on images from a single GAN and reported that, with suitable pre-processing and data augmentation, it generalised to other CNN-based generators. Ojha *et al.* [8] observed that detectors trained in this way fail to generalise to newer families of generative models. They proposed classifying images in the feature space of a pre-trained CLIP model that was never trained for real-versus-fake detection.

Tan *et al.* [9] argued that the up-sampling layers used by both GANs and diffusion models leave local correlations between neighbouring pixels. Their Neighbouring Pixel Relationships (NPR) representation subtracts from an image a copy that has been down-sampled and up-sampled again, which removes most scene content and keeps these local traces. A shallow residual network (the first two stages of a ResNet-50) is then trained from scratch on the NPR representation instead of the raw image.

Provenance records

A different approach records an image's origin when it is created. The C2PA specification [3] defines *Content Credentials*: a manifest embedded in the file that lists how the image was made (for example, created by a trained model) and is signed with a certificate, together with a hash of the image data. Such a record is reliable when present and intact, but it depends on the cooperation of the creating software and is removed by many platforms and editors, so it complements rather than replaces pixel-based detection.

2.4 Datasets and Dataset Bias

GenImage [16] is a benchmark of about 1.33 million real images (from ImageNet) and 1.35 million images from eight generators, with tasks for cross-generator and degraded-image classification. Grommelt *et al.* [4] showed that GenImage contains JPEG-compression and image-size biases, and that detectors trained on it learn these biases. They reported that removing the biases improved robustness to JPEG

compression and raised cross-generator performance by more than 11 percentage points for ResNet-50 and Swin-T detectors.

OpenFake [5] is a recent benchmark that pairs real photographs with images from current proprietary and open-source generators. Its test split contains only generators that are absent from its training split, including GPT Image 1.5, Midjourney 7 and Flux.2, together with real images from ImageNet and DOCCI.

Chapter 3

Proposed Work

3.1 Overview

The pipeline has five stages: data collection, normalisation, splitting, model training and evaluation. Every experiment saves its metrics to a shared results folder, from which the tables in this report and the demo application are produced.

3.2 Dataset

We use Tiny GenImage [10], a scaled-down version of GenImage with 28,000 training and 7,000 validation images. Each image is labelled as real or fake and with the model that produced it. The dataset description lists eight generators, but our inventory found no images from SD14; the seven generators present are ADM, BigGAN, GLIDE, Midjourney, SD15, VQDM and Wukong. Real images come from ImageNet. Table 2 summarises the inventory.

The two classes are balanced (14,000 real and 14,000 generated training images), but the dataset carries the same shortcuts found in our pilot study. Every real image is a JPEG and every generated image is a PNG, so the rule "PNG means fake" labels all 35,000 images correctly. Image size is also tied to the source: each generator produces a fixed resolution, so size alone separates BigGAN, Midjourney and the real images from the rest.

Source	Train	Val.	Format	Median size (px)	Median file (KB)
Real (ImageNet)	14,000	3,500	JPEG	500×375	109
ADM	2,000	500	PNG	256×256	117
BigGAN	2,000	500	PNG	128×128	29
GLIDE	2,000	500	PNG	256×256	75
Midjourney	2,000	500	PNG	1024×1024	1,211
SD15	2,000	500	PNG	512×512	448
VQDM	2,000	500	PNG	256×256	109
Wukong	2,000	500	PNG	512×512	454

Table 2: Inventory of Tiny GenImage.

3.3 Bias Control

The inventory showed that file format, resolution and colour mode each reveal the source of an image. We therefore rebuilt the dataset so that real and generated images differ only in their pixel content.

1. **Filtering.** BigGAN was removed because its 128×128 images cannot be cropped to the 224×224 input size without resizing. We also removed the 280 greyscale real images and real images whose shorter side is below 224 pixels. ADM's alpha channel was dropped by converting every image to three-channel RGB. In total, 31,108 of 35,000 images remained.

2. **Cropping.** Every image was centre-cropped to 224×224 pixels. We crop rather than resize, because resizing alters the fine pixel-level traces that detectors rely on. The crop offset is rounded down to a multiple of 16 pixels, so that the 8×8 JPEG block grid (and the 16×16 chroma blocks) of the real images stays aligned with the grid of the generated images.

3. **Compression matching.** All real images are JPEG files and all generated images are lossless PNG files. Following the approach of Grommelt *et al.* [4], we estimated the JPEG quality factor of each real image by matching its luminance quantisation table against the standard tables scaled for qualities 1

to 100. Each generated image was then compressed once with a quality factor sampled from the distribution of real training images. After this step, every image in both classes has been JPEG-compressed exactly once.

4. **Splitting and balancing.** The dataset's training split was divided into our training (85%) and validation (15%) sets, stratified by source. The dataset's own validation split was kept as an untouched test set. Within each set, real images were sub-sampled to match the number of generated images.

All real images used standard JPEG quantisation tables. Their quality factor is concentrated at 96 (12,298 of 17,500 images, 70%), with smaller groups at 75, 80, 90, 92 and 85. Table 3 gives the final split sizes and confirms that the median quality factor is the same (96) for both classes in every split.

Split	**Real**	**Generated**	**Per generator**	**JPEG quality (mean / median)**	
				Real	**Generated**
Train	10,200	10,200	1,700	92.4 / 96	92.3 / 96
Validation	1,800	1,800	300	92.4 / 96	92.6 / 96
Test	3,000	3,000	500	94.1 / 96	92.4 / 96

Table 3: Processed dataset: six generators (ADM, GLIDE, Midjourney, SD15, VQDM, Wukong) and real ImageNet images, 224×224 pixels.

Two limitations remain. First, the generated test images were compressed with qualities drawn from the training distribution, while the real test images keep their own, slightly higher, qualities (mean 94.1 versus 92.4). Second, because generators produce different resolutions, a fixed 224×224 crop covers most of a 256×256 ADM image but only a small region of a 1024×1024 Midjourney image (Figure 1). The apparent zoom level therefore still varies with the source.

3.4 Detection Methods

- **M1, trained CNN:** a ResNet-50 pre-trained on ImageNet and fine-tuned for real-versus-fake classification, following the approach of Wang *et al.* [14].

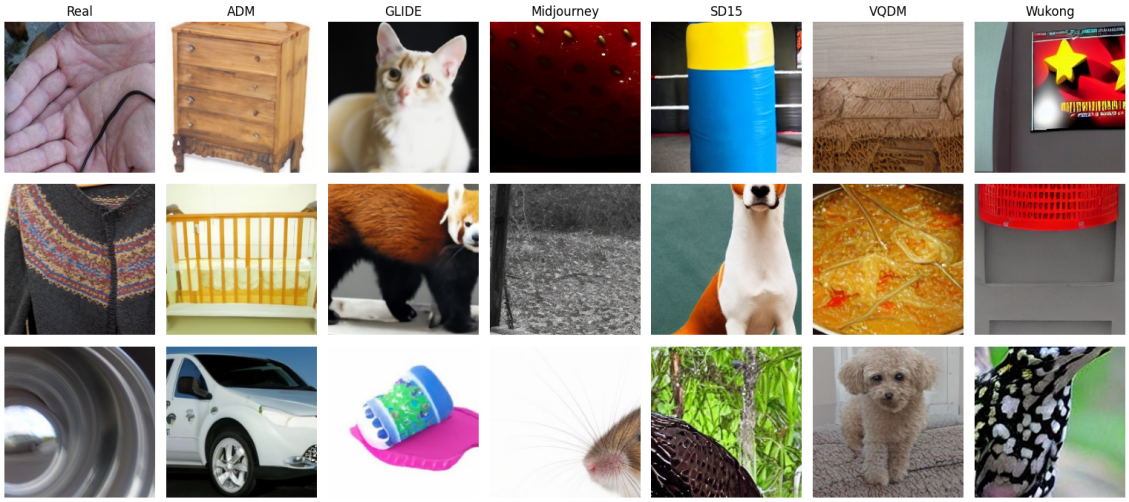


Figure 1: Processed test images, three per source.

- **M2, frozen CLIP features:** image features from a pre-trained CLIP model with a linear classifier on top, following Ojha *et al.* [8].
- **M3, NPR:** the neighbouring pixel relationship representation of Tan *et al.* [9], fed to a shallow ResNet (the first two stages of ResNet-50) trained from scratch, as in the authors' released code.
- **M4, combined detector:** the output scores of M2 and M3 averaged, so that a semantic detector and a pixel-trace detector contribute equally to one decision. M4 needs no additional training.

The three published methods represent three stages of the field: a detector trained on raw pixels (2020), a detector built on a large pre-trained vision model (2023), and a detector built on a forensic residual (2024).

3.5 Training Procedure

All trained detectors share one training procedure. Images are scaled to $[0, 1]$, normalised with the ImageNet channel means and standard deviations, and flipped horizontally at random during training. The network outputs a single score per

image, which is passed through a sigmoid to give the probability that the image is generated. Training minimises the binary cross-entropy loss with the AdamW optimiser (learning rate 10^{-4}, weight decay 10^{-4}), batches of 64 images, mixed-precision arithmetic and a fixed random seed (42), for 5 epochs. After every epoch the model is evaluated on the validation set, and the checkpoint with the lowest validation loss is kept. The test set is used once, with that checkpoint.

3.6 Evaluation Protocol

We report four measures on the test set:

- **Accuracy:** the fraction of images classified correctly when an image is called generated if its probability is at least 0.5.
- **ROC-AUC:** the probability that a randomly chosen generated image receives a higher score than a randomly chosen real image. It does not depend on the 0.5 threshold, so it separates how well a detector ranks images from where its threshold lies.
- **Average precision (AP):** the area under the precision–recall curve, commonly reported in the detection literature.
- **Per-source accuracy:** accuracy computed separately for real images and for each generator, so that a weak generator is not hidden by the overall average.

Misclassified images are inspected in the error analysis.

3.7 Web Application

The project website lets a user upload an image (JPEG, PNG, WebP or iPhone HEIC) and shows the verdict together with the evidence behind it. An upload is analysed in two steps:

1. **Provenance.** The file's Content Credentials (C2PA manifest) are read and verified against the official C2PA trust list (Section 4.13). A valid signed

record stating that a trained model created the image decides the verdict "AI-generated"; a trusted record of a camera capture decides "real photograph". Unsigned AI labels in the metadata (the IPTC digital source type, Stable Diffusion or ComfyUI generation settings) also mark an image as generated. Camera EXIF tags are shown but never used, because they are easy to edit.

2. **Pixels.** If no usable record exists, the image is covered by an evenly spaced grid of tiles of 224×224 pixels at its original resolution (E8; 4×4 by default, 3×3 or 5×5 selectable, Section 4.17), each tile is scored by the CLIP-based detector of E10, and the scores are averaged. The score is rescaled so that the calibrated threshold maps to 50%, and the tile scores are drawn on the image. A score between 40% and 60% is reported as uncertain, with the side it leans to (Section 4.18).

Three further functions support demonstrations. A *WhatsApp simulation* re-saves the uploaded file as in E9 and E10 (long side at most 1600 pixels, JPEG quality 80, metadata removed), analyses the copy and shows both results side by side, so that the loss of the Content Credentials and the change of the pixel score can be seen directly. For one of our ChatGPT images, the simulated copy (204 KB, pixel score 70.8%) closely matched the copy we had actually sent through WhatsApp (199 KB, 70.8%). Six sample images from our own test can be analysed with one click, and while a public link is open the page shows it as a QR code, so that a visitor can analyse a photo from their own phone. The backend is written in Python with FastAPI and runs on our own computer during demonstrations, at about 5–9 seconds per image on a laptop CPU (of which the detector itself takes 3.1 seconds for 16 tiles); a static copy of the page with all experiment results is hosted on Hugging Face Spaces and connects to the backend through a temporary public link. Uploaded images are analysed in memory and not stored. The trained detectors are published in a Hugging Face model repository.

Chapter 4

Experimental Setup and Results Analysis

4.1 Setup

All experiments run on Google Colab with an NVIDIA T4 GPU, using Python, PyTorch, Pillow, NumPy and pandas. Data and results are stored on Google Drive.

4.2 Pilot Study: File-Level Shortcuts (E0)

We collected five real photographs (Apple iPhone 17, HEIC) and five images generated with ChatGPT (PNG). Table 4 shows that the two classes differ on every file-level property.

Property	**Real (5)**	**AI-generated (5)**
Format	HEIF	PNG
Resolution (W×H)	4284×5712	512×512 or 768×512
File size	2,404–5,631 KB	329–524 KB
EXIF tags	11	0
Embedded colour profile	Yes	No

Table 4: File properties of the pilot dataset.

A *shortcut rule* labels an image as AI-generated from one file property, without reading any pixel values. Accuracy is the fraction of images labelled correctly.

4.3 Normalisation (E0b)

Each image was rotated upright using its EXIF orientation, converted to RGB, centre-cropped to 512×512 pixels, rebuilt from its raw pixel array and saved as PNG. A first attempt without the rebuilding step leaked the iPhone colour profile into all real images, which created a new shortcut. Table 5 compares the rules before and after normalisation.

Shortcut rule (predicts "AI" if ...)	**Before (E0)**	**After (E0b)**
File format is PNG	100%	50%
Width under 1,000 pixels	100%	50%
File smaller than 1 MB	100%	50%
No EXIF tags	100%	50%
No embedded information	100%	50%

Table 5: Accuracy of file-level shortcut rules on the pilot set.

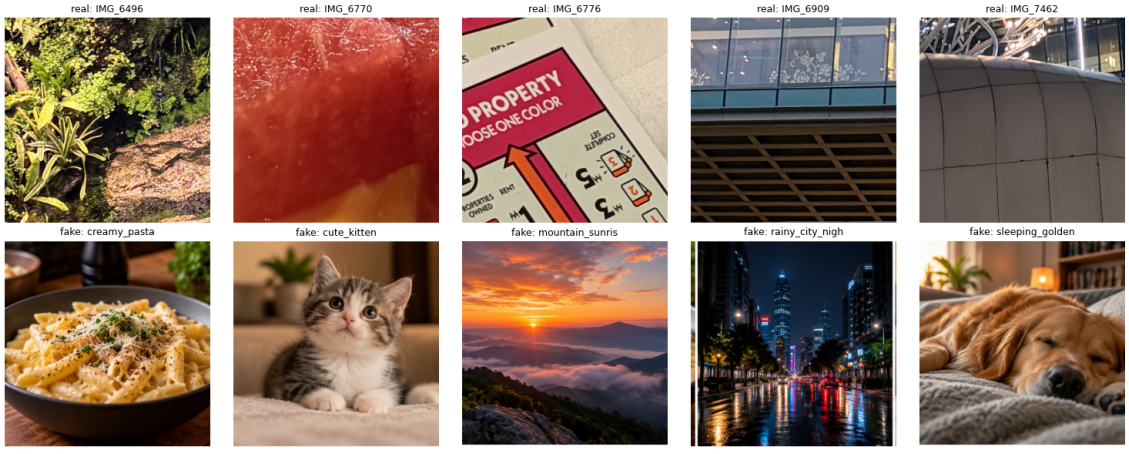


Figure 2: Pilot images after normalisation. Top: real; bottom: AI-generated.

Two biases remain that file cleaning cannot remove. Real crops show close-up

textures, while AI images show centred, well-lit whole scenes (Figure 2). Real pixels also passed through in-camera compression, while AI pixels did not.

4.4 Real versus AI Detection (E1)

E1 trains each detector on all six generators and tests it on the held-out test set of the same generators.

M1: fine-tuned ResNet-50

On the test set, M1 reached an accuracy of 96.5%, a ROC-AUC of 0.994 and an AP of 0.994 (Table 6). Training took about 71 seconds per epoch on a T4 GPU.

Method	**Accuracy**	**ROC-AUC**	**AP**	**Test loss**
M1 (ResNet-50)	0.965	0.994	0.994	0.104
M2 (CLIP + logistic regression)	0.975	0.997	0.996	0.069
M3 (NPR)	0.966	0.994	0.992	0.091
M4 (mean of M2 and M3)	0.988	0.999	–	–

Table 6: E1: test results on the six training generators.

Accuracy differs between sources. Real images were classified correctly 97.2% of the time. Among the generators, GLIDE (99.6%) and SD15 (98.2%) were the easiest to detect, while ADM (91.6%) and Midjourney (93.6%) were the hardest (Table 7).

	Real	**ADM**	**GLIDE**	**MJ**	**SD15**	**VQDM**	**Wukong**
M1	0.972	0.916	0.996	0.936	0.982	0.970	0.954
M2	0.978	0.944	0.992	0.978	0.978	0.974	0.968
M3	0.944	0.976	1.000	0.988	0.988	0.990	0.980

Table 7: E1: per-source test accuracy (for Real, images correctly called real; for generators, images correctly called generated).

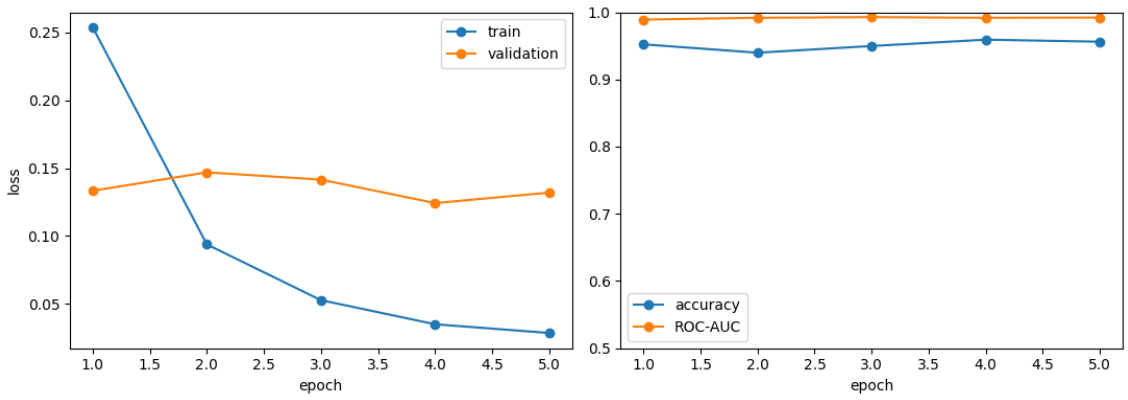


Figure 3: M1 training curves. Training loss keeps falling while validation loss stays near 0.13, a sign of over-fitting after the first epoch.

Figure 3 shows that the training loss fell from 0.25 to 0.03 over five epochs while the validation loss stayed between 0.12 and 0.15. The network continued to fit the training images more closely without improving on unseen ones, so most of what it learned in later epochs did not generalise. Keeping the checkpoint with the lowest validation loss (epoch 4) guards against this.

M2: frozen CLIP features with logistic regression

The CLIP ViT-L/14 image encoder (304 million parameters) was kept frozen, and each image was represented by its 768-dimensional, L2-normalised image embedding. A logistic-regression classifier was trained on these embeddings, with its regularisation strength C chosen from $\{0.01, 0.1, 1, 10, 100\}$ by validation loss. The best value was $C = 100$, although $C = 10$ was almost as good (validation loss 0.082 against 0.081). The trained classifier has only 769 parameters.

M2 reached a test accuracy of 97.5%, a ROC-AUC of 0.997 and an AP of 0.996, slightly above M1 on every measure (Table 6). It was also more even across sources: its lowest per-source accuracy was 94.4% (ADM), against 91.6% for M1, and its accuracy on Midjourney images rose from 93.6% to 97.8%. A frozen general-purpose representation with a linear classifier of 769 parameters therefore matched and slightly exceeded a fully fine-tuned network with about 23.5 million trainable parameters

on the generators seen during training. ADM was the hardest generator for both methods.

M3: NPR

M3 was trained from scratch for 10 epochs with a learning rate of 2×10^{-4} and batches of 32, the settings used in the authors' GenImage example. The network has 1.44 million parameters. Unlike M1, its training and validation losses fell together for most of training (Figure 4). Validation loss was lowest at epoch 8 (0.130), and that checkpoint was used for testing.

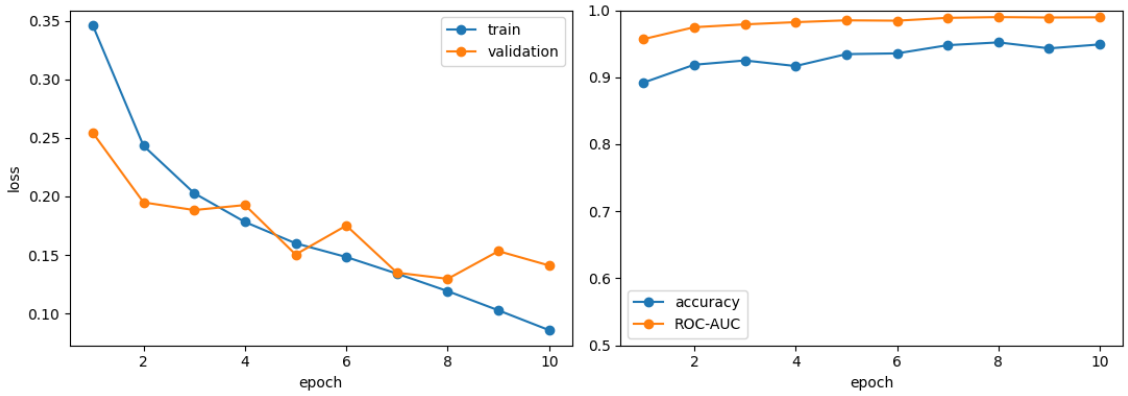


Figure 4: M3 training curves. Training and validation loss decrease together until about epoch 8.

M3 reached a test accuracy of 96.6%, a ROC-AUC of 0.994 and an AP of 0.992, close to M1 overall. Its errors, however, are distributed differently. M3 detected every generator with at least 97.6% accuracy, including ADM, the hardest generator for M1 and M2, and it detected all 500 GLIDE test images. In exchange it misclassified more real photographs: 5.6% of real test images were labelled as generated, compared with 2.8% for M1 and 2.2% for M2. A detector that looks only at local pixel relationships is therefore more sensitive to generated images but also more likely to raise false alarms on real ones.

Comparison

On generators seen during training, all three detectors exceed 96% accuracy and a ROC-AUC of 0.99, so this setting does not separate them strongly. M2 is best overall and the most balanced; M3 is the most sensitive to generated images but has the highest false-alarm rate on real photographs; M1 lies between them. ADM is the hardest generator for M1 and M2 but not for M3. The more informative comparison is on generators the detectors have not seen, which E2 and E6 address.

M4, the average of the M2 and M3 scores, reached 98.8% accuracy and a ROC-AUC of 0.999, the best result in E1, although it required no training. Real-image accuracy rose to 98.1% and the detection rate on generated images to 99.5%: the two methods make different mistakes, so averaging their scores corrects many of them.

4.5 Generalisation to Unseen Generators (E2)

In E2 each detector was trained and validated only on SD15 images and an equal number of real images (3,400 training and 600 validation images), and then tested on the full test set. SD15 is therefore the only generator seen during training; ADM, GLIDE, Midjourney, VQDM and Wukong are unseen. Validation also used only SD15 and real images, so that the unseen generators had no influence on model selection. Because the training set was six times smaller, M1 was trained for 10 epochs and M3 for 30 epochs, each keeping the checkpoint with the lowest validation loss. For M2 the regularisation strength was again selected on the validation set ($C = 100$).

For each generator we report two measures: the detection rate (the fraction of its images labelled generated at the 0.5 threshold) and the ROC-AUC between real test images and that generator's images, which does not depend on the threshold.

The results in Table 8 and Figures 5 and 6 can be summarised in four points.

1. **All detectors work on the seen generator and fail partly on unseen ones.** Every method detected about 97% of SD15 images and kept real-image

	Real	SD15 (seen)	ADM	GLIDE	MJ	VQDM	Wukong	Unseen mean
Correct rate (%)								
M1	97.6	97.8	22.0	61.4	76.8	28.0	84.6	54.6
M2	98.1	97.6	65.4	75.8	95.0	72.4	92.0	**80.1**
M3	96.1	96.4	39.8	69.6	65.2	36.4	87.0	59.6
ROC-AUC (real vs generator)								
M1	–	0.997	0.824	0.956	0.972	0.848	0.983	0.917
M2	–	0.997	0.964	0.981	0.994	0.974	0.993	**0.981**
M3	–	0.991	0.897	0.958	0.934	0.870	0.977	0.927

Table 8: E2: detectors trained on SD15 only. Correct rate is the fraction of real images labelled real, or of a generator's images labelled generated.

accuracy above 96%. On the five unseen generators, the mean detection rate fell to 54.6% for M1 and 59.6% for M3.

2. **M2 (CLIP) generalised best by a wide margin.** Its mean detection rate on unseen generators was 80.1% and its mean ROC-AUC 0.981, against 0.917 for M1 and 0.927 for M3. This agrees with the claim of Ojha *et al.* [8] that a frozen general-purpose representation generalises better than a network fine-tuned on one generator. M2 was best on every unseen generator.

3. **Errors on unseen generators are mainly threshold errors.** For M1, ADM images had a ROC-AUC of 0.824 but only 22% were detected; for GLIDE the ROC-AUC was 0.956 but the detection rate 61.4%. The detectors still scored most unseen generated images higher than real images, but below the 0.5 threshold learned on SD15. The real-image accuracy stayed high, so nearly all errors are missed generated images rather than false alarms.

4. **Generators that resemble SD15 are easier.** Wukong, whose frequency spectrum was close to that of SD15 (Figure 10), was the easiest unseen generator for all three methods (84.6–92.0%). VQDM, whose spectrum showed a



Figure 5: E2: correct rate per source for detectors trained on SD15 only.

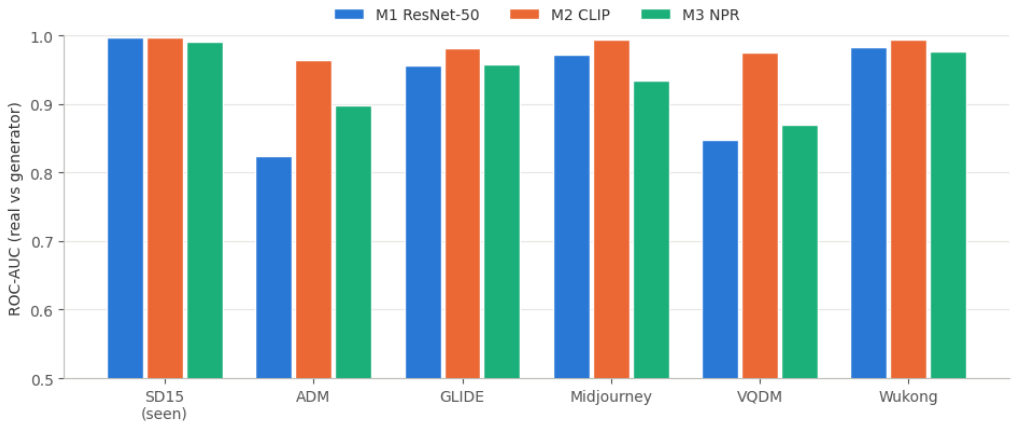


Figure 6: E2: ROC-AUC between real images and each generator.

distinct grid of peaks, and ADM were the hardest. GLIDE, despite its very different spectrum, was of intermediate difficulty, so spectral similarity explains part but not all of the pattern.

M3 also showed unstable validation loss when trained on the smaller SD15-only set (between 0.15 and 0.54 across epochs), and its best checkpoint came from epoch 18.

4.6 Robustness to Image Processing (E3)

Images shared online are rarely the original files. E3 tests the E1 detectors, without retraining, on processed copies of the cleaned test set. Three kinds of processing were applied, each at several strengths: JPEG compression at quality 95, 85, 75, 65 and 50; resizing to 90%, 75%, 60% and 50% of the side length and back to 224×224 pixels (bilinear interpolation); and Gaussian blur with $\sigma = 0.5$, 1.0, 1.5 and 2.0 pixels. Real and generated images received identical processing. The unprocessed results reproduce E1 exactly (M1 0.965, M2 0.975, M3 0.966), and M4 is included.

	None	**JPEG 85**	**JPEG 50**	**Resize 0.75**	**Resize 0.5**	**Blur 1.0**
Accuracy						
M1	0.965	0.884	0.798	**0.826**	0.530	**0.706**
M2	0.975	**0.896**	0.822	0.713	0.527	0.649
M3	0.966	0.601	0.685	0.737	0.517	0.705
M4	**0.988**	0.855	**0.832**	0.728	0.507	0.673
ROC-AUC						
M1	0.994	0.952	**0.913**	0.968	**0.853**	0.940
M2	0.997	0.965	0.904	0.948	0.816	0.916
M3	0.994	0.816	0.764	0.938	0.700	0.913
M4	**0.999**	**0.967**	0.902	**0.974**	0.782	**0.952**

Table 9: E3: selected processing settings (all settings are shown in Figures 7 and 8).

Across all settings (Figures 7 and 8), four patterns appear.

1. **Processing makes real photographs look generated.** Under resizing and blur the detection rate on generated images stayed near 100%, while the fraction of real photographs labelled real collapsed: to between 1% and 6% at half size, and to between 0% and 10% at $\sigma = 1.5$ or more. Accuracy therefore fell to about 50% because the detectors called almost everything generated. Resizing and blur remove fine detail and sensor noise, and a real photograph without them resembles a generated image to these detectors. In practice

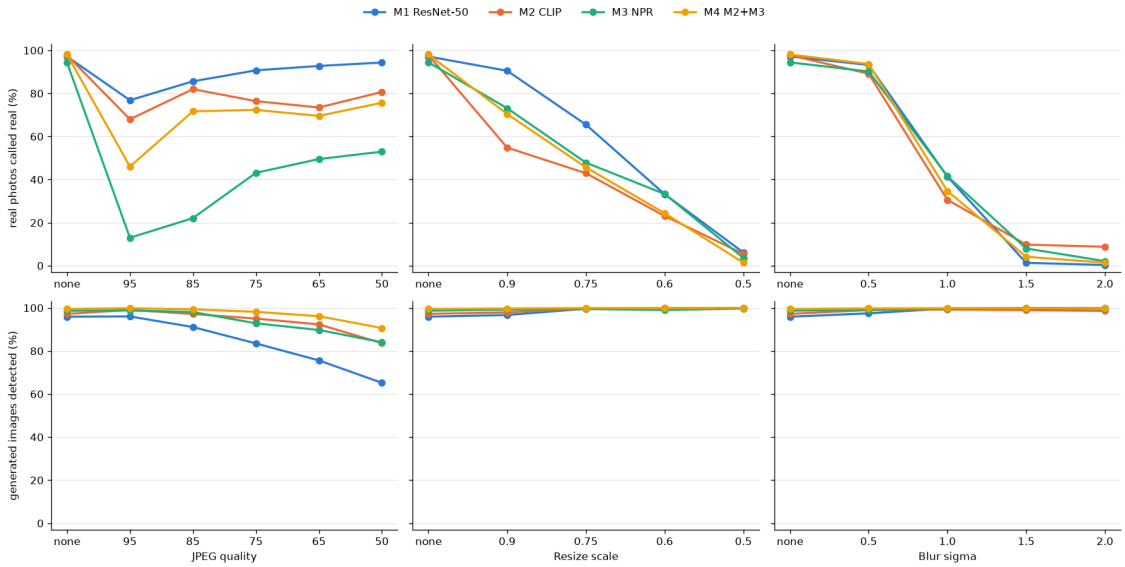


Figure 7: E3: fraction of real photographs labelled real (top) and of generated images labelled generated (bottom) under increasing processing.

this is the more harmful error, since a genuine photograph would be wrongly flagged as fake.

2. **M3 (NPR) is the most sensitive to JPEG.** A single additional compression at quality 95 reduced M3's real-image accuracy from 94.4% to 13.0% and its ROC-AUC to 0.788. NPR measures differences between neighbouring pixels, which JPEG alters in every 8×8 block. For M1, stronger compression had the opposite effect: real-image accuracy recovered while the detection rate on generated images fell to 65.2% at quality 50, so JPEG pushed its scores towards "real". The effect of JPEG did not increase steadily with lower quality; quality 95 caused more false alarms than quality 85 for every method.

3. **No single method is the most robust.** M1 was the most robust to resizing and strong JPEG in accuracy, M2 had the highest accuracy at JPEG quality 85, and M4 had the highest ROC-AUC under blur and moderate resizing. M2, which generalised best to unseen generators in E2, was the most affected by mild resizing (real-image accuracy 54.8% at 90% size).

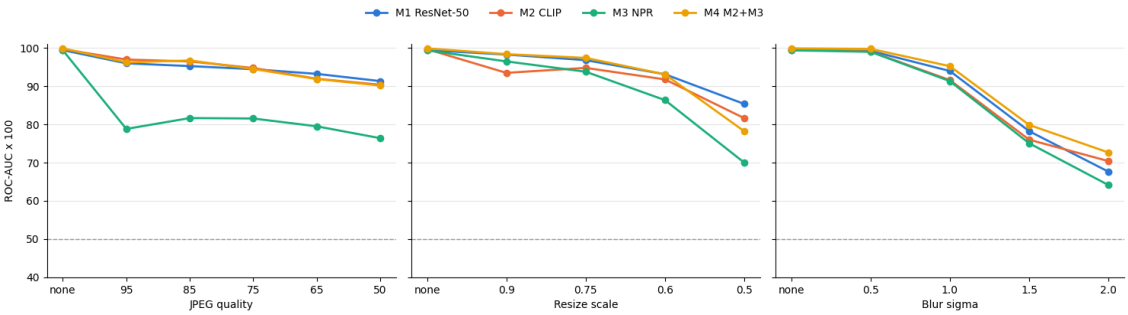


Figure 8: E3: ROC-AUC under increasing processing.

4. **Ranking survives longer than the threshold.** ROC-AUC fell much more slowly than accuracy; at 75% size all methods kept a ROC-AUC above 0.93 while their accuracy was 0.71–0.83. As in E2, the detectors still ranked most generated images above real ones, but processing shifted all scores upwards past the 0.5 threshold.

Wang *et al.* [14] reported that adding blur and JPEG to the training images improves robustness to these operations. Our detectors were trained without such augmentation, which is a likely reason for their sensitivity. Cropping was not tested separately: cropping followed by rescaling to the input size is an enlargement, which overlaps with the resizing test.

4.7 Generator Attribution (E4)

E4 asks which of the seven sources (real or one of the six generators) produced an image. Real images were sub-sampled in every split to the size of one generator class (1,700 per class for training, 300 for validation and 500 for testing), so all seven classes are equally represented and random guessing scores 14.3%. Two attribution models were trained:

- **A1:** the frozen CLIP features of M2 with a seven-class logistic regression. The regularisation strength was selected on the validation set ($C = 10$).

- **A2:** ResNet-50 fine-tuned as in M1, with seven outputs, a softmax and the cross-entropy loss. Validation loss was lowest at epoch 3.

	Real	ADM	GLIDE	MJ	SD15	VQDM	Wukong	Acc.	F1
A1 CLIP	**95.8**	78.4	90.8	83.6	67.8	77.8	66.0	80.0	0.800
A2 ResNet	92.2	**89.6**	**95.8**	**90.8**	**75.8**	**90.4**	**68.4**	**86.1**	**0.861**

Table 10: E4: per-class recall (%), accuracy (%) and macro-F1 (F1) for seven-way attribution. Chance level is 14.3%.

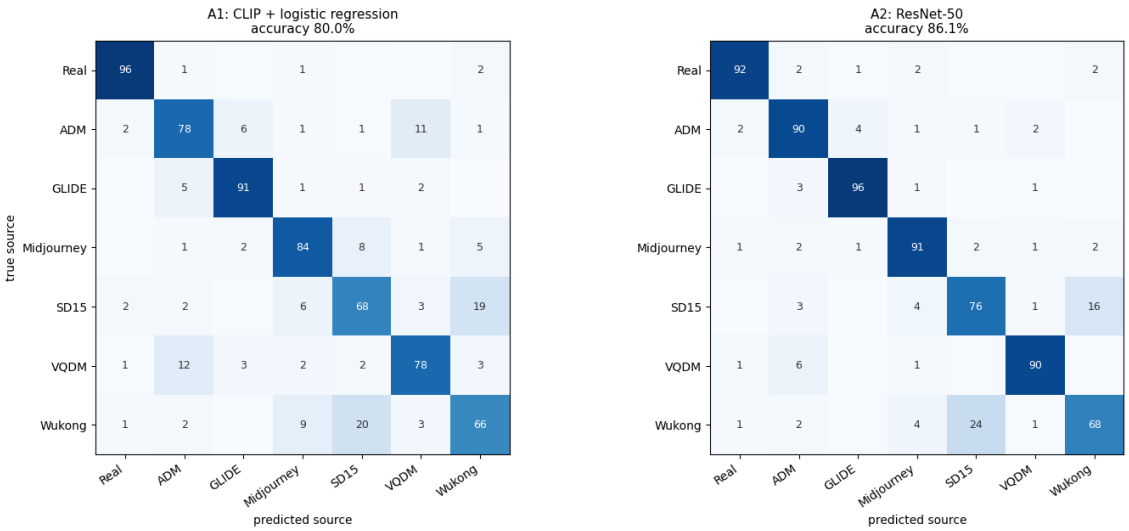


Figure 9: E4: confusion matrices (% of each true source). Rows are the true source, columns the predicted source.

Both models were far above the chance level of 14.3% (Table 10, Figure 9): A1 reached 80.0% and A2 86.1%. Each generator therefore leaves traces that differ not only from real photographs but also from the other generators.

Unlike in E2, the fine-tuned network was the better model here. A2 beat A1 on every generator, by up to 12.6 points (VQDM), and A1 was ahead only on real images (95.8% against 92.2%). CLIP was the best at recognising unseen generators, but telling known generators apart depends on low-level traces that a fine-tuned network learns and that frozen semantic features capture less well.

Most of the remaining errors fall between SD15 and Wukong: 16–19% of SD15 images were attributed to Wukong, and 20–24% of Wukong images to SD15. Their averaged frequency spectra were also the most similar (Figure 10), and Wukong was the easiest unseen generator for a detector trained on SD15 in E2, so three separate measurements agree that the two generators leave similar fingerprints. A1 also confused ADM and VQDM (11–12% in each direction), which A2 largely separated.

4.8 Visualising Generator Fingerprints

A single image does not reveal a generator's traces because the scene dominates. Following the logic of camera fingerprinting, we removed the scene and averaged over many images, so that content cancels out and only consistent traces remain.

Frequency spectra of the noise residual

For each test image we converted it to greyscale, subtracted a copy blurred with a Gaussian filter ($\sigma = 1$) to obtain the noise residual, and computed the magnitude of its two-dimensional Fourier transform. Figure 10 shows the logarithm of the average spectrum over 500 images per source (top) and its difference from the real-image average (bottom). The horizontal and vertical lines through the centre appear in every source, including real images; they come from the image borders rather than from any generator.

The difference maps are mostly blue towards the edges: generated images carry less high-frequency energy than real photographs and are smoother at the finest scale. GLIDE has the most distinctive spectrum. Its energy is confined to a diamond-shaped central region and the corners are almost empty, which is consistent with an image produced at a lower resolution and then enlarged; GLIDE was also the easiest generator for M1 (99.6% accuracy). VQDM and, more faintly, Midjourney show a regular grid of bright points, the periodic peaks expected from repeated up-sampling, while SD15 and Wukong have spectra close to each other and to those of real images.

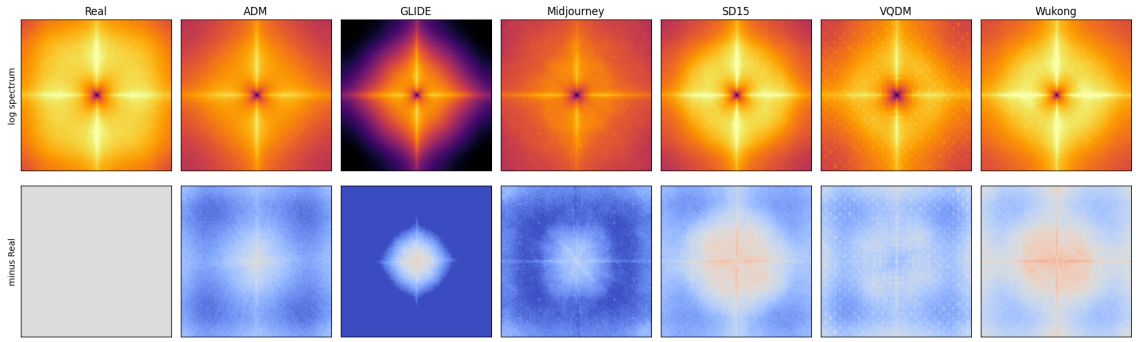


Figure 10: Average log spectrum of the noise residual per source (top) and difference from real images (bottom; blue = weaker than real, red = stronger).

NPR residuals

Figure 11 shows a 64×64 region of one test image per source and the magnitude of its NPR residual. In the real image the residual follows the edges of the scene. In the SD15 example, a nearly uniform blue region still produces a dense, regular residual pattern, which is a trace of the generation process rather than of the content. Table 11 lists the mean absolute NPR value over 500 images per source. Every generator has a lower value than real images, meaning neighbouring pixels are more alike in generated images. GLIDE has the lowest value (3.6 against 9.6 for real images), while SD15 and Wukong are closest to real images.

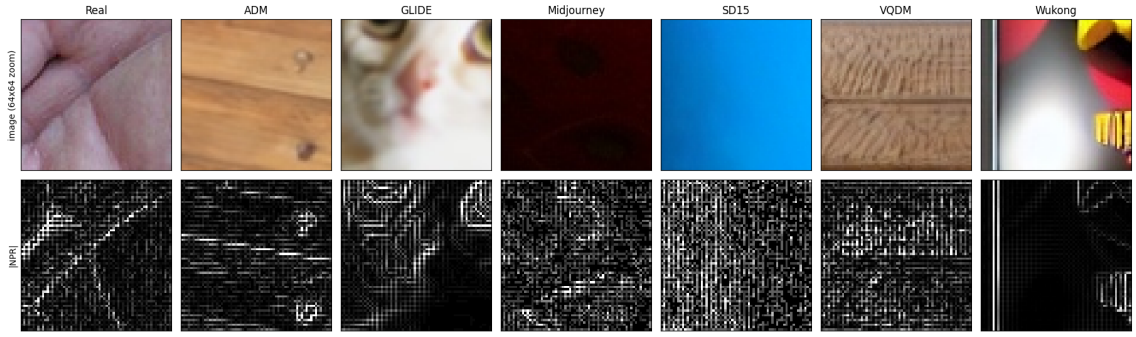


Figure 11: Image regions (64×64, top) and the magnitude of their NPR residual (bottom).

	Real	ADM	GLIDE	MJ	SD15	VQDM	Wukong
Mean |NPR|	9.63	5.79	3.60	5.05	8.55	7.15	9.03

Table 11: Mean absolute NPR residual per source (pixel units, 500 images each).

4.9 Effect of De-biasing (E5)

E5 measures how much of a detector's accuracy can come from the compression shortcut described in Section 3.3 rather than from the image content. We built a second, *uncleaned* copy of the dataset that skips one step of the bias control: generated images are cropped but not JPEG-compressed. Everything else is identical, including the selected images, the crop positions and the splits; the real images are the same pixel arrays in both copies. On the 3,000 generated test images, the JPEG step changed pixel values by 2.03 levels out of 255 on average, a difference that is not visible to a human viewer.



Figure 12: A 64×64 patch of an SD15 test image without (left) and with (centre) the JPEG step. The amplified difference (right) is concentrated along edges and fine detail, where JPEG discards high-frequency information.

A new M1 detector was trained on the uncleaned training set with the same settings as in E1 (5 epochs, lowest validation loss at epoch 4). It was then tested on both test sets, together with the E1 M1 detector, which was trained on the cleaned data. The cleaned–cleaned result reproduces E1 exactly (96.5%), which confirms

that both runs used the same test images.

Trained on	Tested on	Accuracy	ROC-AUC	AP
Uncleaned	Uncleaned	0.988	0.999	0.999
Uncleaned	Cleaned	0.789	0.936	0.942
Cleaned (E1)	Uncleaned	0.870	0.969	0.968
Cleaned (E1)	Cleaned	0.965	0.994	0.994

Table 12: E5: M1 trained on uncleaned or cleaned data, tested on both versions of the test set.

Trained	Tested	Real	ADM	GLIDE	MJ	SD15	VQDM	Wukong
Uncleaned	Uncleaned	98.5	98.6	99.6	97.8	99.4	99.8	99.2
Uncleaned	Cleaned	98.5	41.8	71.6	62.4	63.4	55.4	61.8
Cleaned	Uncleaned	97.2	51.2	97.2	93.6	85.2	68.8	65.0
Cleaned	Cleaned	97.2	91.6	99.6	93.6	98.2	97.0	95.4

Table 13: E5: correct rate (%) per source.

The model trained on uncleaned data scored 98.8% on its own test set, higher than any detector in E1 (Table 12). When the same generated images were JPEG-compressed like the real ones, its accuracy fell to 78.9%, a drop of 19.9 percentage points, and on the fair test set it was 17.6 points below the model trained on cleaned data. The uncleaned benchmark therefore overstates what the detector can do.

The per-source results show what it had learned (Table 13, Figure 13). Its accuracy on real images was unchanged (98.5%), but its detection rate fell for every generator, to between 41.8% (ADM) and 71.6% (GLIDE): compressing a generated image was enough to make it look real. In effect the model had learned that JPEG traces mean "real", as Grommelt *et al.* [4] found on the full GenImage dataset. The shortcut was not all it learned, however. Its ROC-AUC on the cleaned test set remained 0.936, well above chance, so it had also picked up genuine generator traces; the shortcut mainly pushed their scores below the threshold.



Figure 13: E5: overall accuracy for both models on both test sets (left) and per-source correct rate on the cleaned test set (right). The dashed line marks 50%.

The model trained on cleaned data also lost accuracy on the uncleaned test set (96.5% to 87.0%), mostly on ADM (51.2%), Wukong (65.0%) and VQDM (68.8%). Every generated image it saw during training had been JPEG-compressed, so uncompressed generated images were outside its training distribution. A mismatch in processing therefore hurts in both directions, which motivates the robustness tests in E3.

4.10 Current Generators and Real-World Sharing (E6)

Generators released in 2024–2026

E6 tests the E1 detectors, without retraining, on generators that are two to four years newer than those in GenImage. We used the held-out test split of OpenFake v2 [5], whose generators do not appear in its own training split and whose real images come from DOCCI and ImageNet. Three of its thirteen test files were downloaded. We kept up to 200 images per generator (generators with fewer than 50 usable images were left out) and 300 images from each real source, applied the same rules as before (no greyscale images, at least 224 pixels per side) and the same 16-pixel-aligned 224×224 centre crop. In total 3,091 images were used.

Unlike our cleaned GenImage data, OpenFake was used *as distributed*, without re-compression. Its files differ in format: all Midjourney 7 images are PNG files, 64% of GPT Image 1.5 images are PNG, and most other generators are JPEG files of quality 90–95. DOCCI photographs are 2048×1536 JPEG files of quality 75, while ImageNet photographs are about 500×375 pixels at quality 96. We recorded the format and estimated JPEG quality of every image (Table 14).

Source	**Released**	n	**PNG**	**M1**	**M2**	**M3**	**M4**
DOCCI (real)	–	300	0%	63.3	23.7	47.7	30.7
ImageNet (real)	–	300	0%	95.7	97.7	94.3	98.7
Recraft v3	2024-10	200	0%	6.5	63.5	0.5	9.5
Illustrious (fine-tunes)	2024-12	200	20%	78.0	92.0	69.0	85.0
Sora 2 (video)	2025-03	135	10%	52.6	40.7	40.7	42.2
Midjourney 7	2025-04	200	100%	37.0	72.0	41.0	57.5
Veo 3 (video)	2025-06	200	0%	89.0	90.0	90.0	98.5
Wan 2.5 (video)	2025-09	200	0%	79.5	91.5	87.5	97.0
Nano Banana Pro	2025-11	76	9%	81.6	84.2	80.3	90.8
Z-Image Turbo	2025-11	200	26%	85.0	90.0	84.5	91.5
GPT Image 1.5	2025-12	200	64%	25.5	49.0	29.5	35.0
Flux.2 Klein 9B	2026-01	200	0%	20.0	69.5	90.0	84.5
Seedream 5.0	2026-02	87	10%	52.9	21.8	16.1	12.6
ERNIE-Image	2026-04	77	56%	68.8	59.7	63.6	66.2
ERNIE-Image Turbo	2026-04	154	38%	70.8	57.1	58.4	60.4
GPT Image 2	2026-04	111	20%	13.5	34.2	27.0	25.2
Aurora	unknown	71	100%	88.7	98.6	83.1	98.6
Ideogram 2.0	unknown	70	100%	20.0	18.6	20.0	12.9
Lumina	unknown	110	100%	78.2	76.4	59.1	65.5
All generated images: detected (%)				54.6	68.7	57.0	63.3
All real images: called real (%)				79.5	60.7	71.0	64.7
ROC-AUC, all images				0.751	0.720	0.718	0.738

Table 14: E6: correct rate (%) on OpenFake per source. "Released" is the earliest date in OpenFake's metadata for that generator; PNG is the share of PNG files.

We draw five conclusions from Table 14 and Figures 14 and 15.

1. **Detectors trained on older generators fail on current ones.** ROC-AUC fell from 0.994–0.999 on GenImage to 0.718–0.751 on OpenFake, and the detection rate on generated images from 96–100% to 55–69%. No method stayed above a ROC-AUC of 0.76.

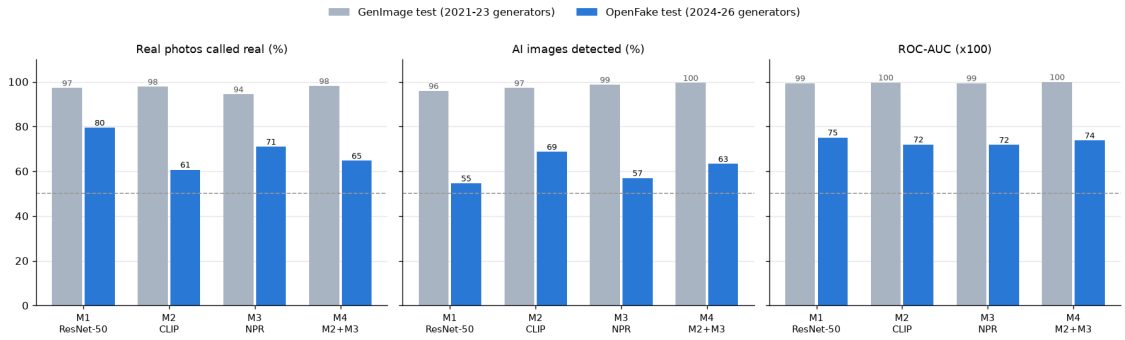


Figure 14: E6: the same detectors on the GenImage test set (E1) and on OpenFake.

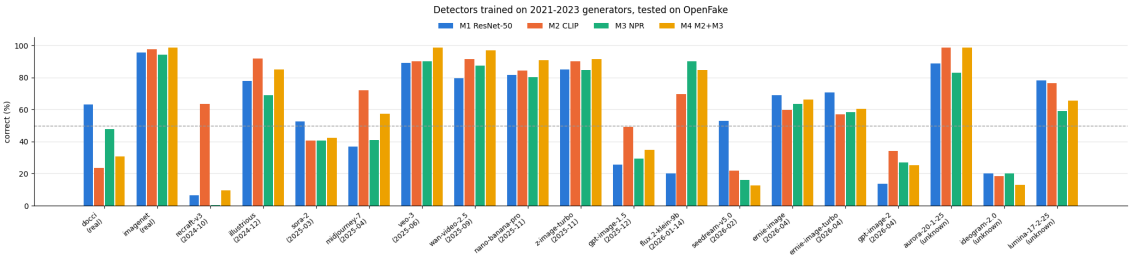


Figure 15: E6: correct rate per OpenFake source, generators ordered by release date.

2. **The two ChatGPT generators are among the hardest.** GPT Image 1.5 was detected at 25.5–49.0% and GPT Image 2 at 13.5–34.2%, with ROC-AUC between 0.49 and 0.63, close to chance. Ideogram 2.0, Seedream 5.0 and, for M1 and M3, Recraft v3 were similarly hard. Frames from video generators (Veo 3, Wan 2.5) and Z-Image Turbo were detected at 80–99%.

3. **Newer is not uniformly harder.** Flux.2 Klein (January 2026) was detected at 90.0% by M3, while Recraft v3 (October 2024) was detected at 0.5% by the same method. Difficulty depends on the generator rather than on its release date. M3 was the only method to detect Flux.2 Klein reliably (ROC-AUC 0.899 against 0.623 for M1).

4. **One real-image source causes many false alarms.** 94–99% of ImageNet photographs were called real, but only 24–63% of DOCCI photographs. DOCCI images are four times larger and more strongly compressed, so a

224×224 crop shows a magnified, smoother region. E3 showed that loss of fine detail makes real photographs look generated to these detectors; we consider this the likely cause but did not test it separately.

5. **PNG files are detected less often than JPEG files of the same generator.** For GPT Image 1.5, M1 detected 16.4% of PNG files and 42.3% of JPEG files; for Illustrious 43.9% and 86.8%; for Z-Image Turbo 73.1% and 89.2%. Our detectors saw only JPEG-compressed generated images during training, and E5 showed that uncompressed generated images are harder for them. M2 was the least affected (Illustrious: 92.7% PNG, 92.1% JPEG).

The evaluation has limitations. Only three of the thirteen OpenFake test files were used, several generators have fewer than 200 images, and OpenFake's release dates vary within one generator, so we report the earliest. Because OpenFake images are larger than GenImage images, the same 224×224 crop covers a smaller part of the scene.

Sharing through WhatsApp

Images are often seen only after being shared through a messaging app. We selected 100 of the OpenFake files: 25 DOCCI and 25 ImageNet photographs and 10 images from each of GPT Image 1.5, GPT Image 2, Midjourney 7, Nano Banana Pro and Flux.2 Klein. They were sent through WhatsApp Desktop as ordinary photos at standard quality, and the received copies were collected from the application's media folder. WhatsApp renamed every file, so each received image was matched to its original by comparing 32×32 greyscale thumbnails with an optimal one-to-one assignment; all 100 were matched.

WhatsApp converted every image to JPEG at an estimated quality of 80 and reduced images larger than 1600 pixels to 1600 pixels on the longer side, which affected the DOCCI photographs and some Nano Banana Pro images. Both versions of each image were cropped and scored in the same way (Figure 16).

Sharing made every detector worse on both classes (Table 15). ROC-AUC fell

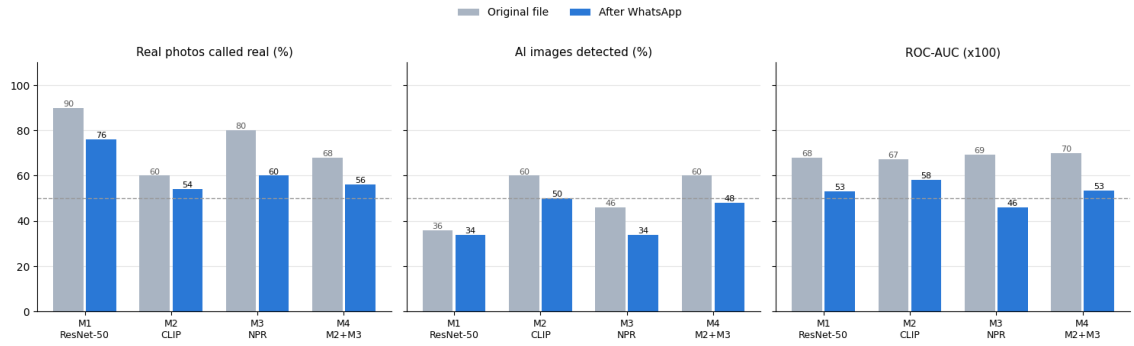


Figure 16: E6: the same 100 images (50 real, 50 generated) before and after sharing through WhatsApp.

	Real called real		**AI detected**		**ROC-AUC**	
	before	after	before	after	before	after
M1	90%	76%	36%	34%	0.680	0.530
M2	60%	54%	60%	50%	0.671	0.580
M3	80%	60%	46%	34%	0.692	0.459
M4	68%	56%	60%	48%	0.701	0.533

Table 15: E6: detector results on the 100 images before and after WhatsApp.

from 0.67–0.70 to 0.46–0.58, close to chance; M3, which depends on neighbouring-pixel differences, fell below 0.5. More real photographs were labelled generated (for example 80% to 60% called real by M3), and fewer generated images were detected (for M3, 46% to 34%). The result combines the two effects measured in E3, re-compression and resizing, in a single everyday operation. With 50 images per class, each percentage has an uncertainty of several points, so per-generator results from this test are not interpreted.

4.11 Training on Current Generators (E7)

E6 showed that detectors trained on 2021–2023 generators miss many images from 2024–2026 generators. E7 asks whether more recent training data closes this gap.

M1 and M2 were trained again, with unchanged settings, on data from the OpenFake *training* split: real photographs from LAION and Pexels and generated images from 72 generators released between 2022 and 2025, including SD 3.5, Flux.1, Midjourney 6, DALL-E 3 and GPT Image 1. Five training files were chosen by evenly spaced position in the file list; in this run that selected one full file and four small ones, giving 3,972 balanced training images, about a fifth of our GenImage training set. One file of the OpenFake validation split (3,552 balanced images) was used for model selection (M1: epoch 2; M2: $C = 10$). The bias control of Section 3.3 was applied: images that were not JPEG files were compressed once at a quality drawn from the real training images, after which the median JPEG quality was 95 for real and 91 for generated training images.

None of the E6 test generators occurs in the OpenFake training split, and its real test photographs come from different sources (DOCCI and ImageNet), so E7 still measures generalisation to unseen generators. The new detectors were evaluated on exactly the images of E6, on the 100 WhatsApp pairs and on our GenImage test set.

		Trained on GenImage			**Trained on OpenFake**		
Test set		Real	AI	AUC	Real	AI	AUC
GenImage	M1	97.2	95.9	**0.994**	50.4	85.0	0.770
	M2	97.8	97.2	**0.997**	81.7	81.8	0.900
OpenFake	M1	79.5	54.6	0.751	64.5	85.3	**0.877**
	M2	60.7	68.7	0.720	88.8	84.8	**0.945**
WhatsApp (after)	M1	76	34	0.530	68	62	**0.669**
	M2	54	50	0.580	86	68	**0.853**

Table 16: E7: real images called real (%), generated images detected (%) and ROC-AUC for detectors trained on GenImage or on OpenFake. The WhatsApp rows use 50 images per class.

Recent training data closed most of the gap (Table 16, Figure 17). On the OpenFake test set, ROC-AUC rose from 0.751 to 0.877 for M1 and from 0.720 to 0.945 for M2, although the test generators were still unseen and the training set was

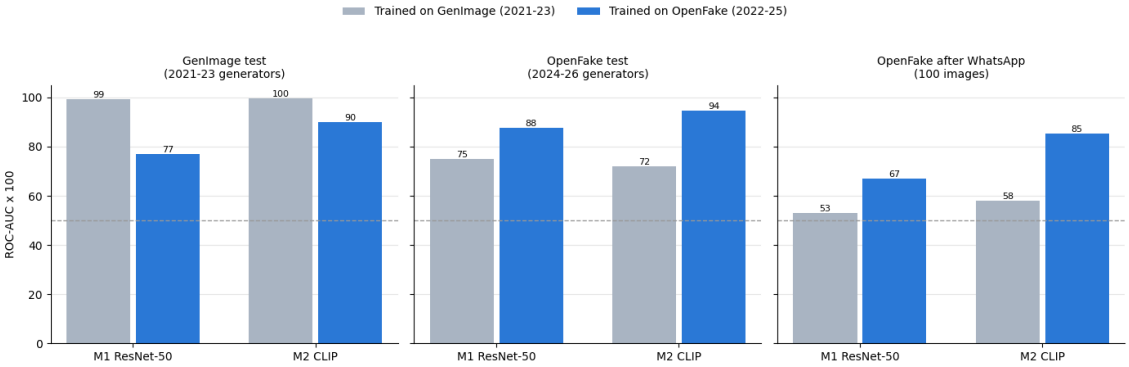


Figure 17: E7: ROC-AUC of detectors trained on GenImage (grey) or on OpenFake (blue).

five times smaller than our GenImage training set. The hardest generators of E6 improved most: for M2, GPT Image 1.5 rose from 49.0% to 80.5% detected, GPT Image 2 from 34.2% to 62.2%, Recraft v3 from 63.5% to 89.5% and Midjourney 7 from 72.0% to 86.0%, and 90.0% of DOCCI photographs were now called real, against 23.7% before. As in E2, the frozen CLIP features gained most and became the best detector.

The gain came at a price on the old generators. On GenImage, ROC-AUC fell to 0.770 (M1) and 0.900 (M2), and the new M1 called only 50.4% of GenImage real images and 47.0% of ImageNet test photographs real: real photographs unlike its LAION and Pexels training photographs were often labelled generated. A detector reflects the real and generated images it was trained on, in both directions.

Robustness to sharing improved but was not solved. After WhatsApp the new M2 kept a ROC-AUC of 0.853, against 0.580 for the GenImage-trained M2, possibly because web images from LAION already carry resizing and compression. The new M1 reached only 0.669, and GPT Image 2 and Seedream 5.0 remained the hardest generators (46–62% detected).

4.12 More Data, Augmentation and Full-Image Analysis (E8)

E8 repeats E7 with five complete OpenFake training files (17,780 balanced training images from 74 generators; 3,446 validation images) and separates two possible improvements:

- **Full data:** the larger training set, with each training image stored as a 256×256 centre crop and a random 224×224 crop taken at every step.
- **Full data + augmentation:** the same, but each training crop is additionally blurred (σ from 0.1 to 2), resized to 50–100% and back, and JPEG-compressed (quality 50–95), each with probability 0.5, following Wang *et al.* [14]. For M2, the classifier is trained on the CLIP features of the clean crops and of two augmented copies of each.

E8 also tests **full-image analysis**: instead of one centre crop, the image is covered by an evenly spaced grid of up to 3×3 tiles of 224×224 pixels at its original resolution, and the tile scores are averaged. Nothing is resized, so pixel traces are preserved.

Test set		**E7**	**Full data**	**+ Augmentation**
OpenFake (centre crop)	M1	0.877	**0.879**	0.832
	M2	0.945	**0.957**	0.941
GenImage (centre crop)	M1	0.770	**0.917**	0.777
	M2	**0.900**	0.881	0.849

Table 17: E8: ROC-AUC of detectors trained on OpenFake data.

More training data helped (Table 17, Figure 18), most of all for M1 on old generators: with 4.5 times more images, its ROC-AUC on GenImage rose from 0.770 to 0.917 and its real-image accuracy there from 50.4% to 89.2%. M2 improved slightly on OpenFake, from 0.945 to 0.957, the best result of the project on unseen 2024–2026 generators.

		Centre crop			Full image		
		AUC	Real	AI	AUC	Real	AI
Before WhatsApp	M1	0.840	88	60	0.871	92	54
	M2	0.927	84	84	**0.953**	94	62
After WhatsApp	M1	0.707	86	50	0.753	98	32
	M2	0.846	80	72	**0.933**	96	32

Table 18: E8 (full data, no augmentation): the 100 WhatsApp images analysed with one centre crop or with the full-image tile grid. Real: real images called real (%); AI: generated images detected (%), both at the threshold 0.5.

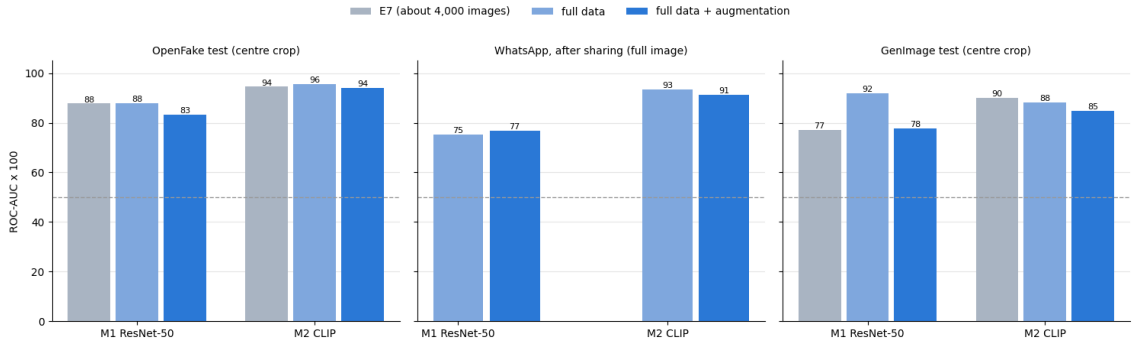


Figure 18: E8: ROC-AUC of the E7 detectors and of the two E8 variants.

Augmentation did not help in our setting. The augmented variants were worse on both test sets and after WhatsApp sharing (M2 on OpenFake: 0.941 against 0.957), although they did improve the hardest generators for M2: GPT Image 2 rose from 58.6% to 70.3% detected, Seedream 5.0 from 66.7% to 75.9% and Sora 2 from 68.1% to 78.5%. One possible reason is that our augmentation was stronger than the processing applied by WhatsApp (quality about 80, at most a reduction to 1600 pixels); we did not test other strengths.

Full-image analysis raised ROC-AUC in all eight comparisons (two detectors, two training variants, before and after WhatsApp), by 0.03 to 0.09; on the WhatsApp images (Table 18) it rose for M2 after sharing from 0.846 to 0.933. At the fixed threshold of 0.5, however, the averaged scores were lower overall: more real images were called real (80% to 96%) but fewer generated images were detected (72% to

32%). Tiles from plain regions of a generated image often look real, and averaging pulls the score down. As in E2, E3 and E6, the ranking was better than the threshold suggests, so a full-image detector needs a threshold chosen for full-image scores, which E9 and E10 provide.

4.13 Provenance Check: Content Credentials

When we tested an early version of our website on images we had generated ourselves, the pixel detector missed both ChatGPT images. Their files, however, carried a different kind of evidence. Both contained a *Content Credentials* record as defined by the C2PA specification [3]: a manifest stored inside the file (in a `caBX` chunk of the PNG) that states how the image was made and is signed with an X.509 certificate. The manifest of each file contained:

- the action `c2pa.created` with the software agent "ChatGPT" (version "gpt-image") and the IPTC digital source type `trainedAlgorithmicMedia`, the standard code for media created by a trained model;
- the signer "OpenAI OpCo, LLC" and a signing time;
- a hash of the image data, so that any change to the pixels after signing can be detected.

We verified both files with the open-source C2PA library (`c2pa-python`) against the official C2PA trust list of certificate authorities. Both were reported as *Trusted*: the signature was valid, the certificate chained to a listed authority and the image hash matched. When we changed a single pixel and kept the manifest, the result changed to *Invalid* with the message that the hashes did not match. The iPhone photograph (HEIC) carried no manifest, only ordinary EXIF tags ("Apple iPhone 17"), and the copies of all images received through WhatsApp carried no metadata at all.

Content Credentials are therefore strong evidence when present, but they are easily lost: WhatsApp, screenshots and most editors remove them, and a missing record says nothing about an image. The two kinds of evidence complement each other, and our website uses both (Section 3.7): a valid signed record decides the

verdict; when there is none, the pixel detector decides. A record that claims a camera capture is accepted only from a signer on the trust list, because anyone can sign a false claim with a self-made certificate.

4.14 A Detector for Current Generators (E9)

E6 to E8 trained only on OpenFake's training split, which leaves out the newest generators so that generalisation to unseen generators can be measured. For the website the goal is different: it should work as well as possible on images that people upload today. E9 therefore trains a *deployment* detector on everything we can spare. OpenFake test files 4 to 12, which no earlier experiment used, contribute 62,145 images (30,386 real from DOCCI and ImageNet, and 31,759 generated, including 3,845 GPT Image 1.5, 319 GPT Image 2, 284 Nano Banana Pro and 190 Aurora images), in addition to the E8 training data. Test file 3 is used to calibrate the threshold and files 0 to 2, the images of E6, for testing.

Each training image is prepared the way the website sees it: up to four tiles from the website's 3×3 grid at original resolution, once as distributed and once after a simulated WhatsApp share (longest side at most 1600 pixels, JPEG quality 80, as measured in E6). The detector is the same as M2: frozen CLIP features and logistic regression with balanced class weights, fitted on 445,112 tiles.

The test generators are now *seen* during training, so E9 measures the detection of known generators on new images, not generalisation.

On OpenFake, E9 was almost perfect (Table 19). After the simulated share it detected 90.0% of GPT Image 2 images (E8: 66.7%), 97.8% of GPT Image 1.5 (71.1%), 97.3% of Nano Banana Pro (67.1%) and 97.2% of Aurora (85.9%). On the real WhatsApp images of E6 it detected 96% of the generated images after sharing, against 58% for E8.

Our own images told a different story. The iPhone photograph received an AI score of 62% and was called AI-generated, with six of its nine tiles above 80%. All real images in OpenFake come from two collections, DOCCI and ImageNet, so the detector had learned what *those* photographs look like; a 24-megapixel phone

Test set	Detector	As distributed AUC	Real	AI	After simulated share AUC	Real	AI
OpenFake files 0–2	E8	0.956	89.3	86.4	0.951	95.1	76.4
	E9	**0.999**	99.3	96.1	**0.998**	99.3	96.0
		Before WhatsApp			**After WhatsApp**		
E6 WhatsApp images	E8	0.953	88	92	0.933	94	58
	E9	**0.998**	98	96	**0.997**	98	96

Table 19: E9: full-image ROC-AUC, real images called real (%) and generated images detected (%), each detector at its own calibrated threshold.

photograph, with its own sharpening and noise reduction, lay outside that range. E10 measures how large this problem is and addresses it.

4.15 Real-World Photographs and Current Generators (E10)

E10 adds four public sources that OpenFake does not cover (Table 20). Phone photographs come from Wikimedia Commons [15], whose "Taken with …" categories are filled from EXIF data: 30 JPEG photographs of 3 to 26 megapixels (normal photo modes, no panoramas) for each of 57 phone models from 2019 to 2026, 47 Android and 10 iPhone models. Unsplash Lite [11] contributes photographs whose EXIF data names a phone maker, mostly older iPhones, and 1,500 photographs from other cameras. Generated images come from ShareGPT-4o-Image [2] (ChatGPT, GPT-4o) and from a public collection of Gemini 2.5 Flash Image (Nano Banana) images [1]. Only CLIP features were kept; no image is redistributed.

The new training tiles (45,312) are few next to the more than 445,000 OpenFake tiles, so they were weighted to provide 30% of the training signal. The threshold was calibrated on the OpenFake calibration images together with the new calibration

[1]Including about 35 photographs from Samsung NX system cameras, which our first filter counted as phones.

Source	Images	Train	Calibration	Test
Commons, Android phones (47 models)	1,409	995	150	264
Commons, iPhones (10 models)	300	184	66	50
Unsplash, phones[1]	858	617	147	94
Unsplash, other cameras	1,500	1,053	222	225
ChatGPT (GPT-4o)	2,000	1,422	278	300
Gemini (Nano Banana)	2,000	1,393	305	302

Table 20: E10: new data sources. Commons and Unsplash were split by photographer, so test photographs come from photographers not seen in training.

images. E8, E9 and E10 were then compared on the held-out test images of every source.

The comparison in Tables 21 and 22 and Figure 19 leads to four observations.

1. **A near-perfect benchmark score hid a real-world failure.** E9 reached a ROC-AUC of 0.999 on OpenFake but called only 44.7% of the Unsplash phone photographs and 28.0% of the other camera photographs real. Over all new test sources its ROC-AUC was 0.921. A detector is only as good as the range of its *real* training images, and two photograph collections are not enough.

2. **Adding real-world photographs fixed it without losing the new generators.** E10 called 98–99% of the phone and camera photographs real, for Android phones and iPhones alike, and still detected 97.3% of the GPT-4o and 98.7% of the Nano Banana images. Its ROC-AUC on the new test sources was 0.998 (0.981 after the simulated share), and on the E6 WhatsApp images 0.992 after sharing (96% of real images called real, 96% of generated images detected).

3. **The result holds across phone brands.** Every brand with at least 30 test photographs had 98–100% of its photographs called real as distributed, and at least 93.5% after the simulated share. Realme was the weakest (81.5%, 27 photographs).

Test images	As distributed			After simulated share		
	E8	E9	E10	E8	E9	E10
Real photographs called real (%)						
Android phones (264)	96.2	86.0	**98.1**	95.1	87.5	**99.2**
iPhones (50)	96.0	84.0	**98.0**	**98.0**	88.0	96.0
Unsplash phones (94)	96.8	44.7	**98.9**	86.2	41.5	**96.8**
Other cameras (225)	94.7	28.0	**98.2**	61.8	12.4	**79.6**
OpenFake real (410)	89.3	99.3	**99.5**	95.1	99.3	**99.8**
Generated images detected (%)						
ChatGPT, GPT-4o (300)	78.0	94.0	**97.3**	53.3	**94.0**	93.0
Gemini, Nano Banana (302)	**99.7**	99.0	98.7	96.7	**99.0**	95.7
OpenFake generated (1,609)	86.4	**96.1**	93.0	76.4	**96.0**	88.6

Table 21: E10: correct decisions on held-out test images, each detector at its own calibrated threshold.

4. **The improvement has a cost on the OpenFake generators.** Detection of OpenFake's generated images fell from 96.1% (E9) to 93.0%, and after the simulated share from 96.0% to 88.6%; the GPT Image 2 rate after sharing fell slightly, from 90.0% to 88.9%. E10 calls more images real, which is the right trade for a public tool, where wrongly accusing a genuine photograph is the more harmful error (E3).

On our own five images, E10 called the iPhone photograph and its WhatsApp copy real (AI scores 28% and 24%) and gave both ChatGPT PNG files scores above the threshold (57% and 63%), but the ChatGPT image received through WhatsApp was still missed (23%).

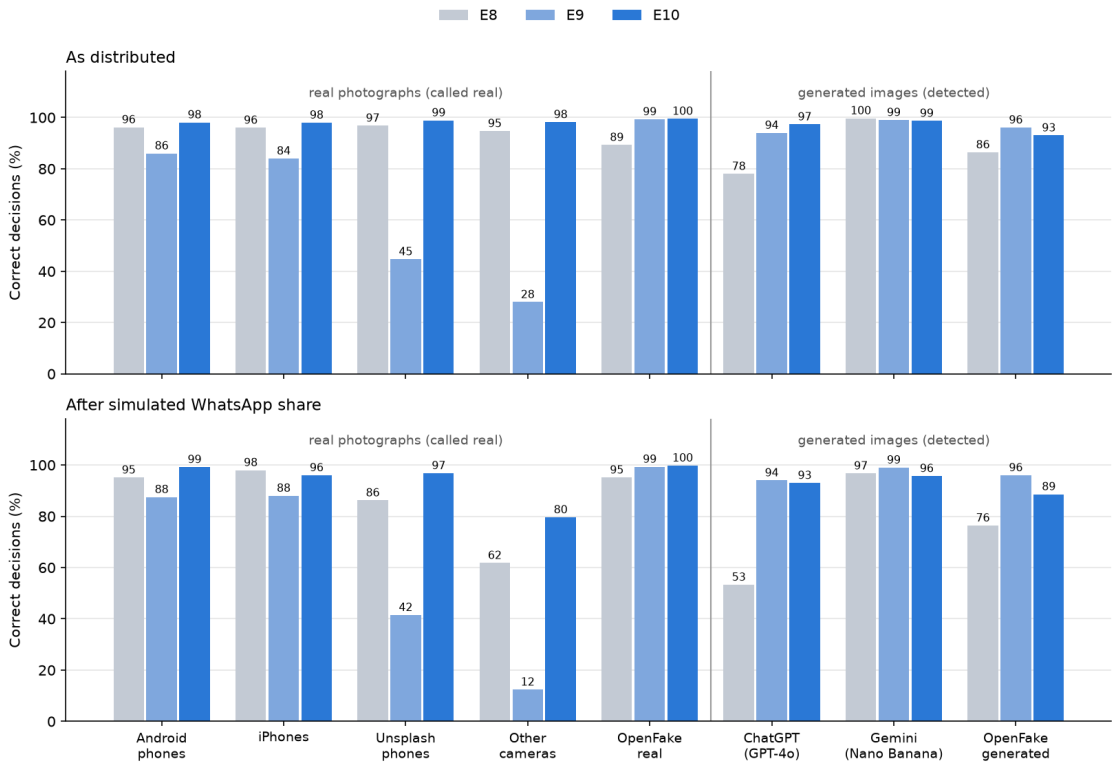


Figure 19: E10: correct decisions of E8, E9 and E10 per test source, as distributed (top) and after a simulated WhatsApp share (bottom).

4.16 Test on Our Own Images from ChatGPT, Gemini and Grok

As a final check we generated new images ourselves in October 2026, after all training was finished: 4 with ChatGPT, 8 with Gemini and 4 with Grok, half in each app on a phone and half on a desktop computer, most of them photorealistic scenes (rooms, people, streets, landscapes, an animal) and a few stylised ones. Each image was tested in four forms: the original download, a screenshot of it, the original sent through WhatsApp, and the screenshot sent through WhatsApp, 62 files in total (one Gemini screenshot was not taken). All files were analysed by the website exactly as a user would upload them.

Brand	Test photos	As distributed	After share
Samsung	71	100.0	100.0
Xiaomi, Redmi, Poco	61	100.0	100.0
iPhone	50	98.0	96.0
OnePlus	38	100.0	100.0
Google Pixel	31	100.0	93.5
Vivo	30	100.0	100.0
Realme	27	81.5	100.0
Oppo, Huawei, Infinix	6	100.0	100.0

Table 22: E10: Commons phone photographs called real (%) per brand.

App	Original	Screenshot	WhatsApp	Both	Total
ChatGPT	4/4	3/4	4/4	3/4	14/16
Gemini	8/8	5/7	5/8	5/7	23/30
Grok	4/4	4/4	4/4	4/4	16/16
Total	16/16	12/15	13/16	12/15	53/62

Table 23: Our own generated images called AI-generated by the website (final verdict). Both: a screenshot sent through WhatsApp.

Table 23 shows that 53 of the 62 files (85%) were called AI-generated. Every original download was identified, most of them by their Content Credentials. All ChatGPT originals carried a trusted manifest signed by OpenAI, and all Gemini originals one signed by "Google LLC", with the source types `trainedAlgorithmicMedia` and `composite` (phone images also `algorithmicallyEnhanced`). The two Grok images saved on the desktop carried a manifest naming "Grok Imagine" as the creating software, but signed with a self-signed certificate marked "local use only", so the signature was valid while the signer could not be verified; the Grok images saved on the phone carried none. Screenshots and WhatsApp copies never kept a manifest.

Without credentials, the pixel detector caught 50 of the 62 files (81%). All 16 Grok files scored above 83%, including two photorealistic images of people, and 14

of 16 ChatGPT files were detected. The weakness was photorealistic outdoor scenes from Gemini: ten of the twelve pixel-level misses came from three Gemini images, a street with cafés, a mountain lake at sunset and a lion in a savanna, whose originals scored 34–47%, while Gemini's indoor scenes scored 67–99%. Many misses were close to the threshold (41–50% for nine of them). With the 16-tile grid that the website uses by default since the tile test of Section 4.17, 57 of the 62 files (92%) were called AI-generated: 16/16 originals, 13/15 screenshots, 14/16 WhatsApp copies and 14/15 screenshots sent through WhatsApp (ChatGPT 15/16, Gemini 26/30, Grok 16/16). All files in this test were generated, so it measures detection only; real photographs were tested in E10 and in Section 4.17.

4.17 Number of Tiles

The website covers an image with a grid of 224×224 tiles at full resolution instead of shrinking the whole image to the detector's input size, because shrinking removes the fine detail in which generator traces lie (E3). A 3×3 grid, however, covers only about a third of a typical image. We therefore compared grids of 1 (the centre crop) to 36 tiles on 206 images: the 64 generated files of Section 4.16 (our 62 and two earlier ChatGPT images) and 142 real files, namely 70 photographs from 14 phone models that were not used for training (from Wikimedia Commons), each as downloaded and after a simulated WhatsApp share, and two of our own iPhone images.

Going from 9 to 16 tiles raised the share of generated files detected from 81.2% to 90.6%, with one more false alarm among the 142 real files (Table 24); more than 16 tiles brought no further gain. The mean score of the real files hardly changed (0.135 for 9 tiles, 0.136 for 16), so the calibrated threshold remains valid. A single centre crop detected many generated files but also called 8.5% of the real files AI-generated. The website therefore uses 16 tiles by default and lets the user choose 9 (faster) or 25.

The tile scores are combined by their mean. On the same 206 files with 16 tiles

Tiles	ROC-AUC	AI detected	Real as real	CPU time
1 (centre crop)	0.984	93.8	91.5	–
4	0.982	75.0	97.9	–
9	0.992	81.2	98.6	1.8 s
16	0.995	**90.6**	97.9	3.1 s
25	**0.996**	**90.6**	98.6	4.8 s
36	**0.996**	**90.6**	97.9	–

Table 24: Final detector (E10) with different numbers of tiles on 64 generated and 142 real files (%, threshold calibrated for nine tiles). CPU time: the detector alone on a laptop. Large images get fewer tiles where the grid does not fit; 36 tiles averaged 30.

we compared other rules: the mean of the log-odds, the mean of the 8 or 4 most AI-like tiles, the highest tile, and two rules that give less weight to tiles with little detail (such as a plain wall), using the average difference between neighbouring pixels. At the same number of false alarms (3 of 142 real files), the mean detected the most generated files (96.9%, against 92.2–95.3% for the rules based on the highest tiles and 81.2–84.4% for the detail-based rules), and it had the highest ROC-AUC together with the log-odds mean and the most AI-like half (0.995–0.996). Rules based on the highest tiles mainly shifted the threshold: they detected every generated file but called up to 44% of the real files AI-generated. Plain regions therefore carry useful evidence, and the mean was kept.

4.18 Uncertain Results

Most of the remaining errors were close calls: of the nine wrong decisions with 16 tiles in Section 4.17, six had a calibrated score between 40% and 60%. The website therefore reports such scores as “not sure” and names the side the score leans to, instead of giving a confident verdict. The band of ten points on either side of the decision line was fixed as a round value and not tuned; it was evaluated on the same 206 files as the number of tiles, so the result is optimistic.

	Correct	Not sure	Wrong
Generated (64), without band	58	–	6
Generated (64), with band	48	14	2
Real (142), without band	139	–	3
Real (142), with band	137	4	1

Table 25: Decisions of the final detector (16 tiles) with and without the uncertain band of 40–60%.

With the band, 18 of the 206 files (8.7%) were reported as uncertain, and the confident decisions were correct for 185 of 188 files (98.4%) instead of 197 of 206 (95.6%) (Table 25). The cost is that 12 files that would have been decided correctly are also reported as uncertain, although with the correct lean.

4.19 Attribution of Current Generators (E11)

E4 attributed images to six generators from 2021 to 2023. E11 trains an attribution model for current generators, grouped into families that users know: ChatGPT (GPT-4o, GPT Image 1, 1.5 and 2, DALL-E 3), Gemini (Nano Banana, Nano Banana Pro, Imagen), Grok (Aurora, Grok 2 Image, Grok Imagine), Midjourney, Flux, Seedream, Ideogram, Recraft, other generators and real photographs. It is a ten-class logistic regression on the CLIP features saved for E8 to E10, with at most 40,000 training tiles per class and balanced class weights; the family of an image is the one with the highest average tile probability.

On the held-out test images, 67% of generated images were attributed to the correct family, six times the chance level (Table 26, Figure 20). Families with one distinctive generator, such as Grok (Aurora only), Recraft and Seedream, were recognised best; Other AI, which mixes dozens of generators, worst. Within Gemini, Nano Banana Pro was attributed correctly for only 34% of its images.

On our own 62 files, however, the model named the right application for only 5 (ChatGPT 0 of 16, Gemini 4 of 30, Grok 1 of 16), fewer than chance; ChatGPT images were mostly attributed to Midjourney, Ideogram or Gemini. The training

Family	Test images	As distributed	After share
ChatGPT	480	70.6	84.4
Gemini	375	72.8	78.9
Grok	71	100.0	100.0
Midjourney	90	90.0	87.8
Flux	90	91.1	93.3
Seedream	84	91.7	90.5
Ideogram	68	66.2	80.9
Recraft	152	97.4	86.8
Other AI	801	45.9	48.1
All generated images	2,211	67.1	71.6

Table 26: E11: generated test images attributed to the correct family (%, nine families; chance 11%).

images of each family come from a few datasets and model versions, and the images produced by the applications today differ from them, as E9 showed for real photographs. Attribution therefore worked within the benchmark but did not transfer to the applications themselves, and the website does not show it. For files that keep their Content Credentials, the signed manifest names the application reliably (Section 4.16).

4.20 Fingerprint Map (E12)

The detectors of E8 to E11 work on CLIP features, 768 numbers per tile. E12 draws these features as a map. Each test image of E10 and E11 is represented by the average of its tile features; t-SNE [12] places the images in two dimensions so that images with similar features lie close together. We used up to 300 images per generator family and 600 real photographs (1,803 images), as distributed and after the simulated WhatsApp share. Distances between distant groups and the size of a group carry little meaning in a t-SNE map, so the map is backed by a measure computed on the full features: for each image, the share of its 10 nearest neighbours

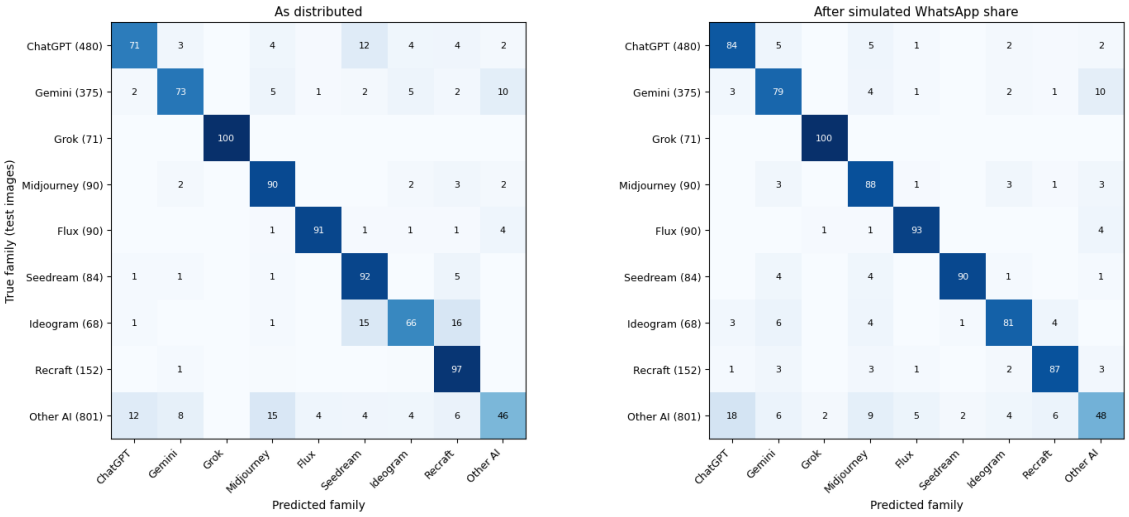


Figure 20: E11: attribution of generated test images to families (% of each true family).

(cosine distance) that belong to the same group, which we call *neighbour agreement*.

Real photographs and generated images occupy clearly separate regions (Figure 21): 96% of the neighbours of a real photograph were real and 94% of the neighbours of a generated image were generated, against 35% and 65% by chance. After the simulated share the agreement fell to 89% and 85%, and part of the real photographs moved into the region of the generated images, in line with E3 and E6: compression and resizing make a photograph resemble a generated image.

Among generated images, every family was more often its own neighbour than chance (Table 27, Figure 22). Grok, represented by a single generator (Aurora), formed one compact group; Gemini and ChatGPT formed several groups each, matching their several model versions; Midjourney, Flux and the mixed Other AI group overlapped most. This agrees with E11, which attributed Grok best and Other AI worst. The map does not separate generator traces from image content: real and generated images come from different collections and show different subjects, so part of the separation may reflect content, a limitation shared by all results on these data (E5).

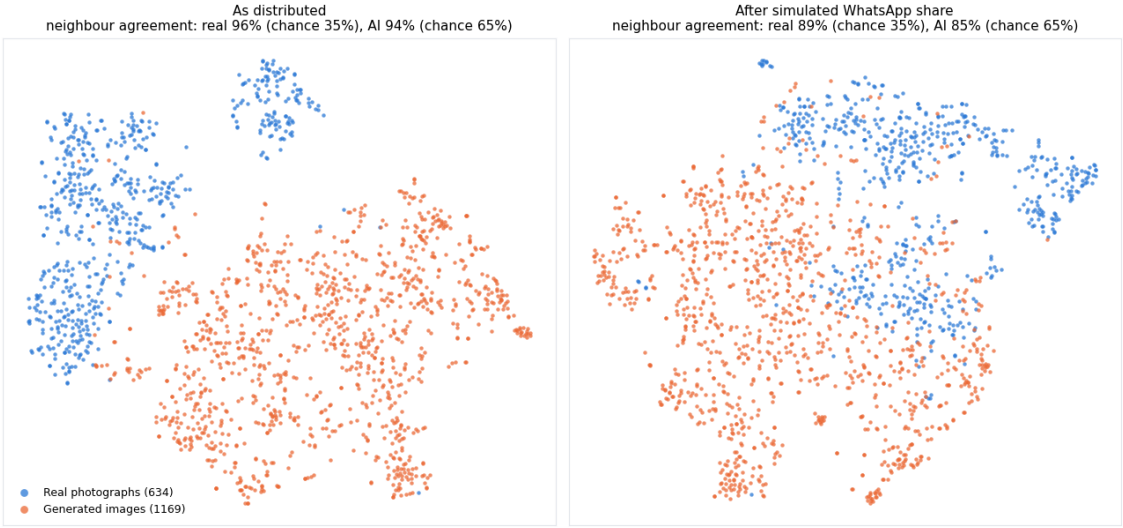


Figure 21: E12: CLIP features of 634 real photographs and 1,169 generated images, mapped with t-SNE, as distributed and after the simulated WhatsApp share.

4.21 Error Analysis

Table 28 collects the cases where the final detector (E10) still fails most often, from the experiments above.

Across all experiments, four patterns explain most of the errors:

1. **Unfamiliar real photographs are called generated.** Resized or blurred photographs (E3), DOCCI photographs for the GenImage-trained detectors (E6) and phone photographs for E9 were all flagged as AI. Smooth images without sensor noise look generated to every detector we trained, and the remedy that worked was more varied real training data (E10).

2. **Generators with few training images are missed.** The weakest generators in E10 had fewer than 300 training images each, and in E6 every generator was unseen. Detection of a generator depends mainly on whether it, or a close relative, was in the training data (E2, E6, E7).

3. **Processing hides generator traces.** JPEG compression, resizing and sharing through WhatsApp lowered the detection of generated images in E3, E6

Family	Images	Chance	As distributed	After share
ChatGPT	303	25.9	66.9	63.2
Gemini	301	25.7	78.0	68.6
Grok	71	6.1	71.4	69.7
Midjourney	90	7.7	45.2	36.9
Flux	90	7.7	43.8	32.6
Other AI	314	26.9	47.5	44.3

Table 27: E12: neighbour agreement among generated images (% of the 10 nearest neighbours from the same family). Seedream, Ideogram and Recraft are counted as Other AI.

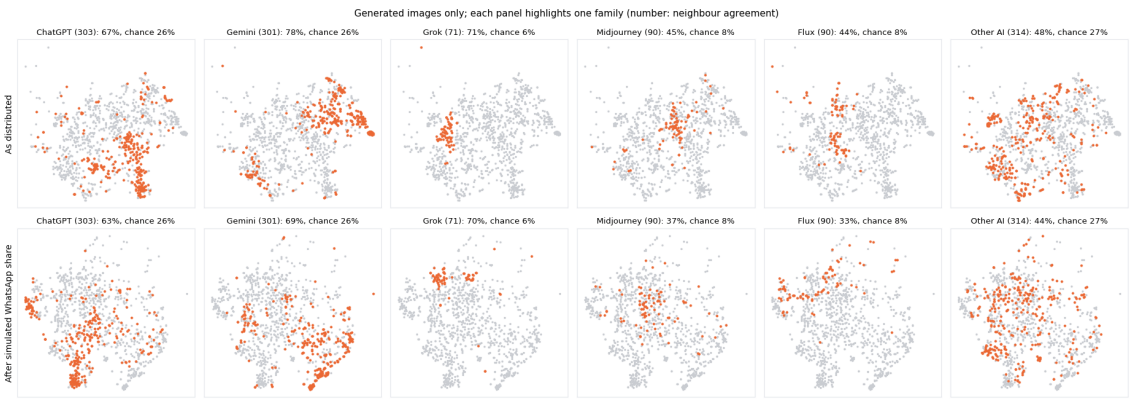


Figure 22: E12: generated images only; each panel highlights one family.

and E8 and, to a lesser degree, in E10.

4. **Scores shift before rankings do.** In E2, E3, E6 and E8, ROC-AUC stayed much higher than accuracy: the detectors still ranked generated images above real ones, but processing or new generators moved all scores across the threshold. Calibrating the threshold on data like the deployment data (E8, E9, E10) recovered most of the accuracy.

Case	Correct (%)	Likely reason
Frames (OpenFake), after share	57.7	a generator with few training images (176), whose images look photographic
Seedream 5.0, as distributed	75.0	263 training images; detected more often after the simulated share (91.7%)
Ideogram 2.0, after share	77.9	187 training images; many show text and graphic design, unlike most training images
Other-camera photographs, after share	79.6	edited, high-quality photographs; resizing removes the sensor noise that marks them as real (E3)
Recraft v2, after share	77.4	few training images (193)
Realme phones, as distributed	81.5	strong in-phone processing; only 27 test photographs
Gemini outdoor scenes, own test	3 images	photorealistic landscapes, a street and an animal; originals scored 34–47%, and most of their copies were also missed

Table 28: Weak spots of the final detector (E10).

Chapter 5

Conclusion and Future Work

5.1 Answers to the Research Questions

RQ1: real versus AI on known generators. Yes. On de-biased data all three detectors reached 96.5–97.5% accuracy and a ROC-AUC of at least 0.994, and their combination 98.8% (E1). This holds only when the data is fair: trained on the uncorrected benchmark, the same network learned that JPEG traces mean "real" and lost 19.9 percentage points on fairly prepared images (E5).

RQ2: unseen generators. Only partly. Trained on one generator, the detectors caught 55–80% of five others (E2), and detectors trained on 2021–2023 generators reached a ROC-AUC of only 0.72–0.75 on 2024–2026 generators; ChatGPT's GPT Image 2 was detected 13.5–34.2% of the time (E6). Frozen CLIP features generalised best. Most errors were threshold errors: unseen generated images were still ranked above real ones but scored below the threshold. Training data from recent generators closed most of the gap (ROC-AUC 0.957, E8).

RQ3: processing. Processing is the main practical weakness. Resizing and blur made real photographs look generated (at half size only 1–6% were called real, E3), and a single WhatsApp share brought every detector close to chance (E6). Recent training data, full-image analysis and a calibrated threshold restored a

ROC-AUC of 0.93 after WhatsApp (E8), and the final detector reached 0.99 (E10), although with the test generators seen in training.

RQ4: attribution. Within a benchmark, yes: a fine-tuned ResNet-50 named the correct one of seven 2021–2023 sources for 86.1% of images (E4), and a CLIP-based model the correct one of nine current generator families for 67% (E11; chance 11%). Each generator also showed a visible, averaged frequency fingerprint. For images made with the applications themselves, no: the E11 model named the right application for only 5 of our 62 files. When Content Credentials survive, they name the application with certainty. In the CLIP features, real and generated images form clearly separate groups, and each family is more often its own neighbour than chance (E12).

Two further lessons came from testing on images we made ourselves. A benchmark score is not a guarantee: a detector with a ROC-AUC of 0.999 on OpenFake called most real phone photographs generated, because all of its real training images came from two photo collections; real-world photographs from 57 phone models fixed this (E10). And pixels are not the only evidence: images downloaded from ChatGPT carry signed Content Credentials that identify them with certainty, but sharing removes them, so a pixel detector remains necessary.

5.2 Limitations

- The final detector (E9, E10) was trained on the generators it was tested on; its results measure detection of known generators on new images, not generalisation, which E2 and E6 measure.
- Some current generators had few training images (319 for GPT Image 2, 190 for Grok's Aurora), and new generators appear every few months; E6 shows how quickly a detector becomes outdated.
- WhatsApp was tested with 100 real image pairs and otherwise simulated by resizing and JPEG compression; other platforms process images differently.

- Attribution of current generators (E11) did not transfer from the benchmark to images made with the applications themselves.
- Each configuration was trained once with a fixed random seed, so small differences between methods may not be significant.
- Our own test (62 files from 16 generated images) is small, and it contains no new real photographs; photorealistic outdoor scenes from Gemini were the main weakness.

5.3 Future Work

- Continual updating: collect images from new generators as they are released and retrain, which with frozen CLIP features takes minutes once the features are computed.
- Attribution that transfers to the applications: training on images collected from ChatGPT, Gemini and Grok themselves rather than from datasets, and testing on images from other people.
- A larger test set collected by ourselves, with images generated and photographed by many people and shared through several platforms.
- Combining evidence: Content Credentials, invisible watermarks where detection tools become available, and pixel analysis.
- Robustness to deliberate attacks, such as adding noise or re-photographing a screen, which this project did not test.

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